

# Computing Written Test for Interview



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Please answer the following Computing subject knowledge questions. Time allowed is 30 minutes.

## Question 1

1a) State the denary representation of the binary number 10010111

$$2^0 + 2^1 + 2^2 + 2^4 + 2^7 = 1 + 2 + 4 + 16 + 128 = 151$$

.....[1]

1b) State the hexadecimal representation of the denary number 125. You must show your working.

.....

.....[2]

125 = 7D<sub>16</sub>

base 16 : 0 1 2 3 4 5 6 7 8 9 A B C D E F

16 | 125  
112  
13

## Question 2

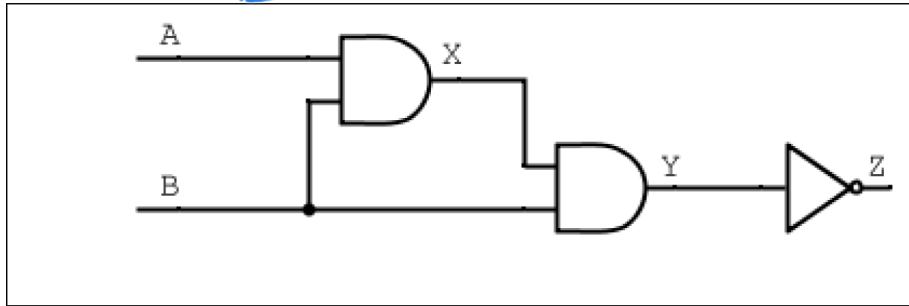
a) Complete the truth table for the OR logic gate:

| A | B | A OR B |
|---|---|--------|
| 0 | 0 | 0      |
| 0 | 1 | 1      |
| 1 | 0 | 1      |
| 1 | 1 | 1      |

(1 mark)

b) Complete the truth table for the following logic circuit

*AND Gate* *Do Inversion*



| A | B | X | Y | Z |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 0 |

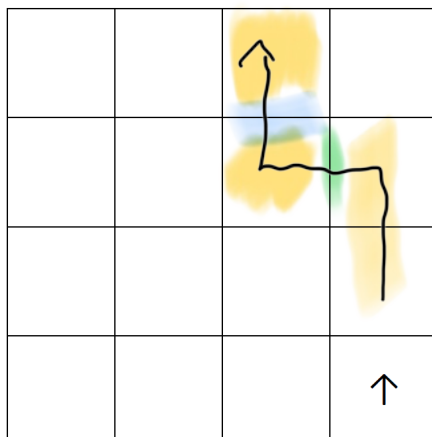
(3 marks)

### Question 3

- a) Draw the path of the robot through the grid below if the following program is executed (the robot starts in the square marked by the  $\uparrow$  facing in the direction of the arrow).

```

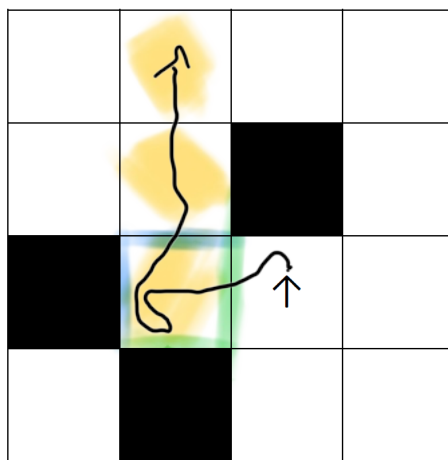
Forward(2)
TurnLeft()
Forward(1)
TurnRight()
Forward(1)
  
```



(3 marks)

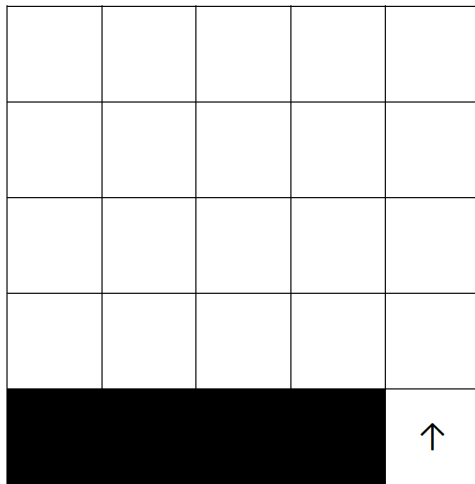
b) Draw the path of the robot through the grid below if the following program is executed (the robot starts in the square marked by the ↑ facing in the direction of the arrow). If a square is black then it contains an object.

```
WHILE ObjectAhead() = true
  TurnLeft()
  IF ObjectAhead() = true THEN
    TurnRight()
    TurnRight()
  ENDIF
  Forward(1)
ENDWHILE
Forward(1)
```

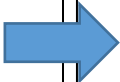


(3 marks)

C A robot needs to visit every square in the following grid that does not contain an object:  
The objects are shown as black squares.



Complete the algorithm by writing the following instructions in the correct places (you will need to use each instruction exactly once):

|                                                                                         |                                                                                     |                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>Forward(distance) distance ← distance - 1 distance ← 4 TurnLeft() TurnLeft()</pre> |  | <pre> distance ← 4 WHILE distance &gt; 0     forward (distance)     Turn Left ( )     Forward(distance)     Turn Left ( )     distance ← distance - 1 ENDWHILE </pre> |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|

(4 marks)

#### Question 4

Heath is researching how long, to the nearest minute, each student in his class spends playing computer games in week (Monday to Friday). He is storing the data in a 2D array. Figure 1 shows part of the array, with 4 students.

|                  | Students |     |    |    |    |
|------------------|----------|-----|----|----|----|
| Days of the week |          | 0   | 1  | 2  | 3  |
|                  | 0        | 60  | 30 | 45 | 0  |
|                  | 1        | 180 | 60 | 0  | 60 |
|                  | 2        | 200 | 30 | 0  | 20 |
|                  | 3        | 60  | 10 | 15 | 15 |
|                  | 4        | 100 | 35 | 30 | 45 |

4 a) Explain why Heath is using an array to store the data.

Arrays are multi-dimensional and allow for storage/access of the minute values based on the two dimensions Student and Day [2]

4 b) i) Identify a datatype that could be used to store the number of minutes in this array.

integer [1]

4b) ii) State why this data type is the most appropriate

uses the least amount of memory to store 0 - 200. No decimals required, not close to overflow. [1]

4c) Heath wants to output the number of minutes student 3 played computer games on Wednesday (day 2).

He writes the code: `print (hoursPlayed[3,2])` . The output is 20.

(i) Write the code to output the number of minutes student 0 played computer games on Wednesday

`print (hoursPlayed[0,2])` [1]

(ii) State the output if Heath runs the code:

`print (hoursPlayed[2,1])`

0 [1]

### Question 5

Zuhair wants to create a Local Area Network (LAN) for himself and his family, in his home.

5(a) Describe what is meant by a LAN.

A Local Area Network is a grouping of <sup>closely located</sup> devices as opposed to a Wide Area Network which covers much more distance / smaller LANs (2 marks)

5(b) TCP/IP uses packet switching.

Explain what is meant by packet switching.

Packet switching involves the breaking down of data into smaller chunks (packets) that include metadata on how they can be recombined following being sent over the network and arriving in any order. This allows for multiple routes to be used and accounts for possible missing packets.

(4 marks)

Written Task – Describe what makes an engaging lesson (Please use full sentences and at least two paragraphs)

An engaging lesson should appropriately account for differences in previous student understanding. For many reasons, students don't all come to the class having had the same experiences of education over the years and it is important for an engaging lesson to be designed to be accessible to all students in the class.

An engaging lesson in my mind involves an exciting and intuitive hook to introduce the lesson. Where possible, lessons should link with students' other subjects, popular cultural references relevant to the topic. Expert questioning can keep the students engaged and discussing/answering between themselves.

The preparation of the lesson is very important. Whether it's unplugged or digital, students should be well informed of the learning outcomes and how this builds on previous knowledge.

To measure the engagement, formative assessment is important for teachers to gain insight into students and students to get important feedback from the teacher.

The lesson might involve group work, and should definitely have a strong conclusion to emphasise/revise the key learning points.